Problem Challenge 2: Rearrange a LinkedList (medium)

Problem Statement

Given the head of a Singly LinkedList, write a method to modify the LinkedList such that the **nodes from the second half of the LinkedList are inserted alternately to the nodes from the first half in reverse order**. So if the LinkedList has nodes `1 -> 2 -> 3 -> 4 -> 5 -> 6 -> null`, your method should return `1 -> 6 -> 2 -> 5 -> 3 -> 4 -> null`.

Your algorithm should use only constant space the input LinkedList should be modified in-place.

Example 1:

```
Input: 2 -> 4 -> 6 -> 8 -> 10 -> 12 -> null
Output: 2 -> 12 -> 4 -> 10 -> 6 -> 8 -> null
```

Example 2:

```
Input: 2 -> 4 -> 6 -> 8 -> 10 -> null
Output: 2 -> 10 -> 4 -> 8 -> 6 -> null
```

Try it yourself

Try solving this question here:

Java

```java
/*class ListNode {
  int val = 0;
  ListNode next;

  ListNode(int value) {
    this.val = value;
  }
}*/

class Solution {

  public ListNode reorder(ListNode head) {
    // TODO: Write your code here

    return head;
  }
}

```

Test Case | Run Code Result | Stdout

[2,4,6,8,10,12]

Use Example Testcases | Run | Submit

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